

Types of SQL Server Installation on Windows Server, with notes, examples, and when to use each type:

✿ Types of SQL Server Installation on Windows Servers

Sno	Installation Type	Description / Notes	Use Case / Example
1	Standalone Installation	Installs a single SQL Server instance on one Windows Server machine. It runs independently and is not part of any cluster or AlwaysOn configuration.	✓ Ideal for small or medium environments where high availability is not mandatory. 📘 Example: Installing SQL Server 2019 Developer Edition on Windows Server 2019 for lab or development use.
2	Failover Clustered Installation (FCI)	SQL Server instance is installed across multiple Windows Servers (cluster nodes) using Windows Server Failover Clustering (WSFC) . Data is stored on shared storage , and SQL runs on one node at a time.	✓ Used for High Availability (HA). 📖 Example: Two-node cluster setup — if Node1 fails, SQL automatically fails over to Node2. 📁 Shared disk: SAN/NAS storage is required.
3	Always On Availability Groups (AG)	Creates multiple replicas of user databases on separate SQL instances. Provides high availability and disaster recovery (HA/DR) without shared storage.	✓ Recommended for mission-critical environments. 🔄 Allows automatic failover, read-only routing, and offloading backups. 📘 Example: Primary replica in DataCenter A and secondary replica in DataCenter B.
4	Standalone Named Instance Installation	Installation of multiple named instances on the same Windows Server. Each instance has a unique name (e.g., MSSQLSERVER, SQL2019DEV, etc.).	✓ Useful for multi-tenant environments, or testing multiple versions on one server. ⚙️ Example: Installing SQL2017 and SQL2019 side by side for comparison.
5	Default Instance Installation	The instance is installed with the default name (MSSQLSERVER). Clients can connect using only the server name without specifying an instance name.	✓ Simple configuration for single SQL instance. 📘 Example: ServerName instead of ServerName\InstanceName during connection.
6	Named Instance Installation	Installed with a specific name different from the default (e.g., ServerName\SQLPROD). Multiple named instances can coexist on the same Windows Server.	✓ Useful for multi-instance scenarios. ⚙️ Example: Server01\FinanceDB and Server01\HRDB.
7	Integration with Active Directory Domain Account Installation	SQL services are configured to run under domain accounts (e.g., DOMAIN\sqlservice). Allows better authentication and centralized security.	✓ Recommended for production deployments. 🔒 Enables Kerberos, AD-based authentication, and service isolation.
8	Local Service Account Installation	SQL services use local accounts like NT SERVICE\MSSQLSERVER or LocalSystem.	⚠️ Suitable for testing, standalone, or non-domain environments.

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			❌ Avoid using in production due to limited network authentication.
9	Cluster Node Add Installation (Add Node to Existing FCI)	Adds a new node to an existing SQL Failover Clustered Instance . The node will act as a standby node ready to take over during failover.	✅ Used during HA expansion or scaling cluster capacity.
10	Passive Node Installation for AlwaysOn	Configures a secondary replica (passive node) for AlwaysOn Availability Groups.	✅ Used for DR (Disaster Recovery) setup with synchronous or asynchronous commit modes.
11	Command-Line (Unattended / Silent) Installation	SQL Server setup is performed via command prompt using configuration files or parameters (e.g., setup.exe /ConfigurationFile=Config.ini).	✅ Used for automation and mass deployments across multiple servers.💡 Common in cloud or DevOps pipelines.
12	SysPrep (Image-Based Installation)	Allows administrators to prepare an SQL Server image (pre-configured binaries) and then complete the setup later with instance-specific settings.	✅ Ideal for VM templates or rapid deployment in enterprise environments. 📘 Example: Cloud-based provisioning of pre-configured SQL images.
13	Upgrade Installation (In-place)	Upgrades an existing SQL Server instance to a newer version (e.g., SQL 2017 → SQL 2019) while keeping configuration and data.	⚠️ Suitable for smaller systems, but risky—rollback is complex. ✅ Recommended to take full backup and test first.
14	Side-by-Side Upgrade Installation	Installs new SQL Server version alongside existing version, then migrates databases and logins manually.	✅ Safest upgrade approach. 📘 Example: Run SQL 2017 and SQL 2019 together, migrate DBs, then uninstall old version.
15	Distributed Replay / Client Tools Only Installation	Installs only management tools or performance testing utilities like SQL Server Management Studio (SSMS) or Distributed Replay Controller/Client .	✅ Useful for DBAs or testers who don't need a full SQL engine installation.
16	Azure Hybrid Installation (SQL IaaS Extension on Azure VM)	SQL Server installed on Windows Server in Azure with SQL IaaS Agent extension for integrated management.	✅ Enables Azure backup, patching, and cost management for SQL on VM.
17	Evaluation Edition Installation	Installs SQL Server with all enterprise features for 180-day trial .	✅ Best for testing enterprise features before purchasing licenses.

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18	Developer Edition Installation	Same features as Enterprise Edition but licensed for non-production/development only .	✓ Ideal for learning, labs, or development servers.

Key Notes

1. Edition Choice Matters:

- *Express* → Free, limited database size (10 GB).
- *Developer* → Full features, non-production use.
- *Standard / Enterprise* → Paid, production-ready.

2. SQL Server Services Installed May Include:

- Database Engine
- SQL Agent
- Analysis Services (SSAS)
- Reporting Services (SSRS)
- Integration Services (SSIS)
- SQL Browser & Full-Text Search

3. Installation Methods:

- **GUI Installation (Setup Wizard)** — simplest for manual installs.
- **Command-line Installation** — for automation.
- **Configuration File Installation** — repeatable, standardized installs.

4. Pre-Installation Requirements:

- Windows updates and prerequisites (e.g., .NET Framework).
- Local or domain service accounts.
- Sufficient disk space for system DBs and backups.
- Admin privileges for installer account.